UNIVERSITY OF LONDON

THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Machine Learning With Python

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Task 01 : Unsupervised Learning

1.1 Introduction

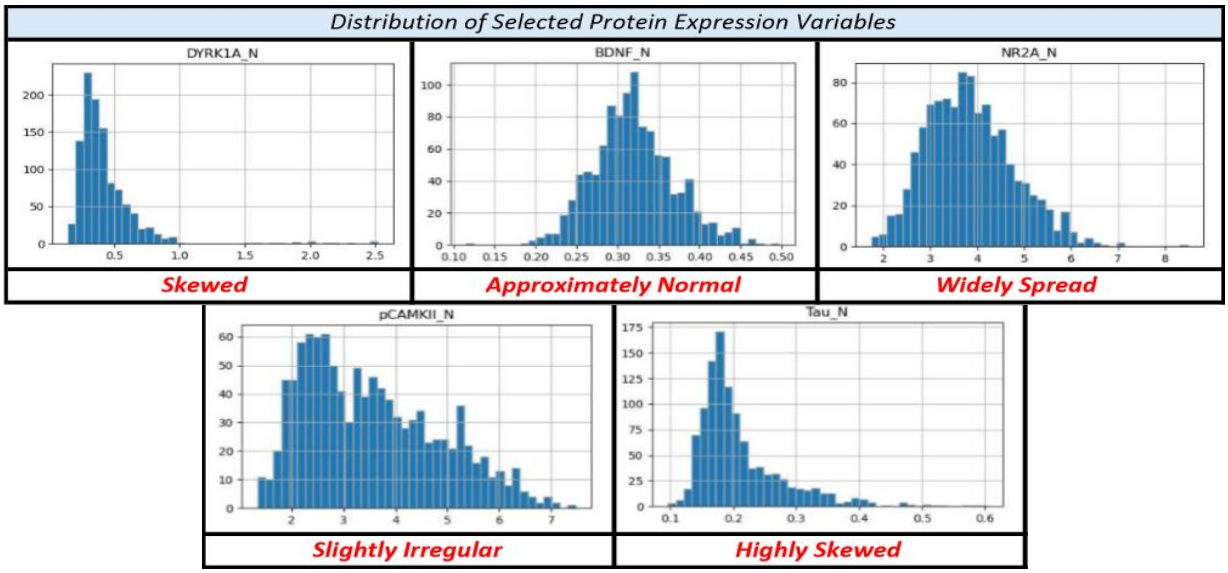
This analysis applies unsupervised learning techniques to analyze the Mice Protein Expression data set obtained from UCI machine learning repository, with the sim to identify underlying patterns of the high dimensional biological data. Machine learning methods such as K-Means, clustering, hierarchical clustering and Principal Component Analysis (PCA) are used to explore potential groupings and interpret variation in the protein expression across experimental conditions.

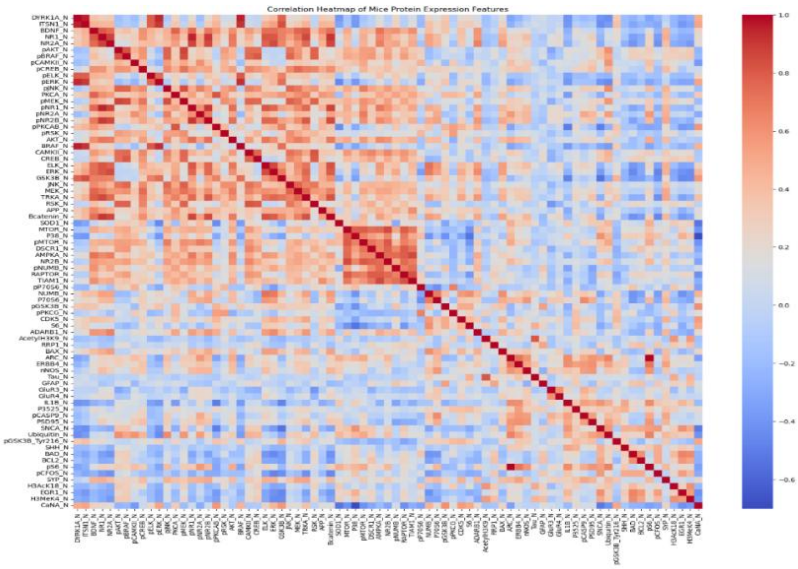
1.2 Research Questions

- Are there any distinct clusters in protein expression data that reflect underlying biological differences between mice?
- How well do clustering methods identify meaningful groupings within the dataset?
- How effectively does PCA reflect the structure of the data set?

1.3 Exploratory Data Analysis

Exploratory data analysis was conducted to examine the distribution and the relationships between the protein expression variables. Histograms of the protein features indicate that most variables follow an approximately normal distribution and while some exhibit slightly skewed distributions, some exhibit noticeable irregular patterns and the presence of outliers. This indicates variability in protein expression levels across samples and suggests that the data may not strictly satisfy the assumption of normality. This variability highlights the complexity of the data set and supports the use of unsupervised learning techniques to identify underlying patterns.



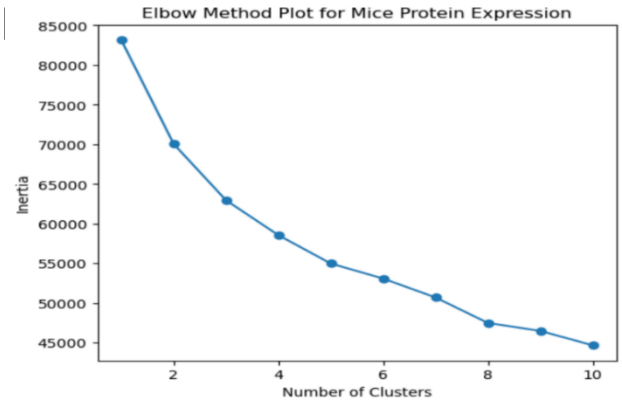


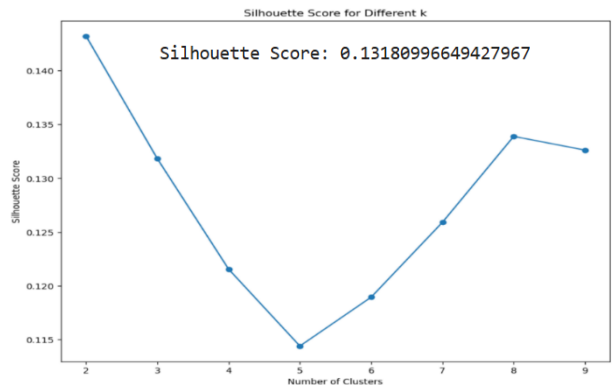
The correlation heatmap reveals a combination of positive and negative relationships among features, with some protein groups showing moderate to strong correlations. This suggests the presence of an underlying structure within the dataset and potential redundancy among certain variables. These relationships further justify the use of dimensionality reduction techniques such as PCA to summarize data effectively.

1.4 Unsupervised Models

1.4.1 K – Means Clustering

K-Means clustering was used to identify potential groupings within the protein expression data. The elbow method was used to determine the optimal number of clusters by analysing the relationship between the number of clusters and within-cluster sum of squares(***inertia***). As the number of clusters increases, the inertia decreases, this indicates an improved model fit.

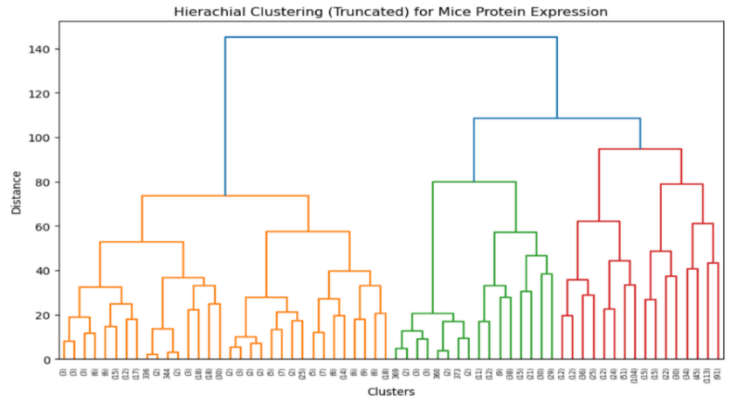




However, the rate of decrease slows beyond a certain point. The elbow point appears around **k = *3***, at which point the inertia begins to level off, suggesting that increasing the number of clusters beyond three provides diminishing returns. Therefore, three clusters were selected as the optimal choice for the K-Means model. In addition to the elbow method, the silhouette score was used to evaluate the quality of clustering.

The silhouette plot shows that the highest scores occur at **k = 2**. Despite this, a value of **k = 3** was selected based on the elbow method and consistency with hierarchical clustering results. The silhouette score generated at **k = 3** is approximately **0.13**, indicating a weak to moderate cluster separation, suggesting that while some structure exists in the data there is a considerable overlap between clusters, and the identified groupings are not highly distinct.

1.4.2 Hierarchical Clustering

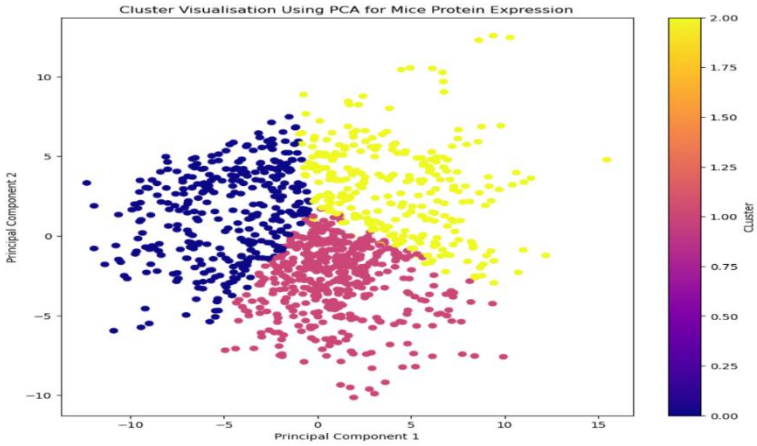


Hierarchical clustering was further applied to examine the structure of the dataset. The dendrogram generated suggests the presence of three main clusters. Larger distances represent greater dissimilarity between clusters, supporting the existence of distinct groupings within the data.

However, the separation between the clusters is not sharply defined at all levels. Some branches merge at relatively lower distances.

This distance indicates the existence of clustering structures although the boundaries between the groups are not strongly produced. Overall, the results generated are consistent with those obtained from K-Means clustering, supporting the choice of three clusters.

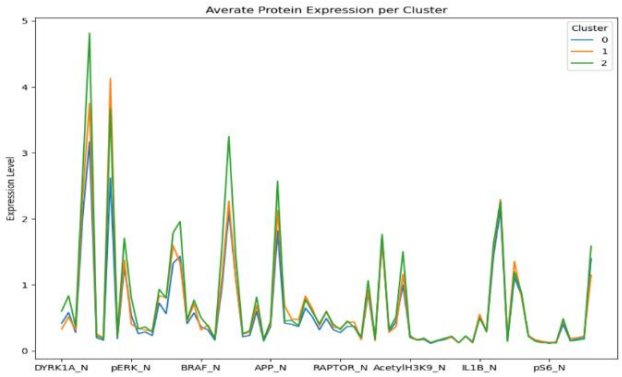
1.4.3 Principal Component Analysis (PCA) Visualization

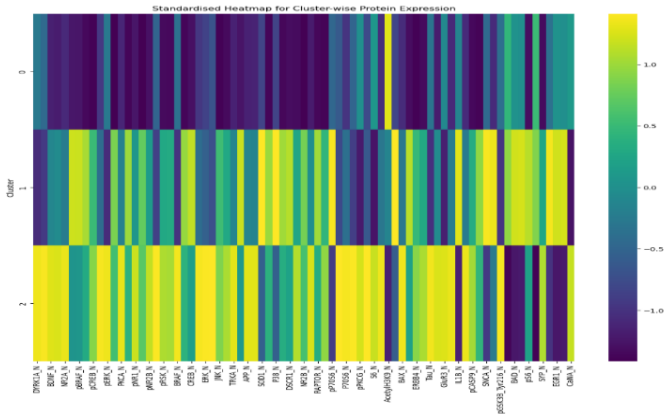


PCA was used to reduce dimensionality of the dataset for visualization. The resulting plot shows clustering structures. However, there is a noticeable overlap between clusters, which indicates a more moderate than stronger separation. This result remains consistent with the low silhouette score, suggesting that the clusters are not highly distinct.

1.4.4 Cluster Interpretation

An average protein expression plot was generated to show the differences in expression levels across clusters. Certain proteins exhibit noticeably higher or lower values in specific groups. This indicates that the clusters capture distinct protein expression patterns even though some similarities remain across clusters. A heatmap of the cluster was visualized to highlight variations in protein expression across clusters, with clear differences in intensity for several proteins. This suggests that the clusters are characterized by different expression profiles, although overlaps between certain clusters are present.





1.5 Conclusion on Unsupervised Learning

Overall, the unsupervised learning on mice protein expression suggests the presence of three clusters within the dataset. While the elbow method and hierarchical analysis support this clustering structure, the relatively low silhouette score and PCA visualization indicate a moderate separation between groups. This implies that although the clustering captures some meaningful patterns in protein expression, the distinctions between clusters are not strongly defined.

Task 02 : Regression

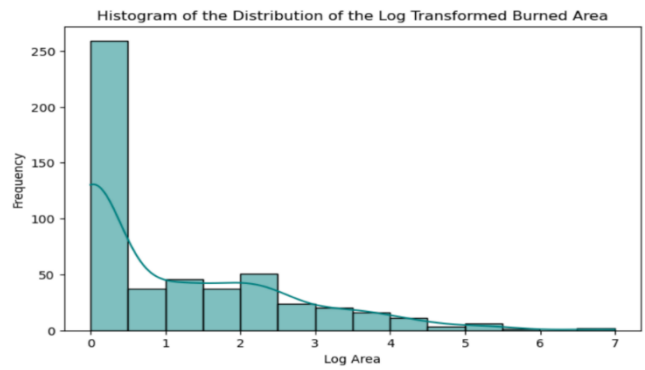
2.1 Introduction

This analysis applies regression techniques to analyze the data on Forest Fires in the northeast region of Portugal, obtained from UCI machine learning repository, with the aim to evaluate the relationship between predictors and fire severity by comparing the performance of different regression models. Machine learning methods such as linear regression, ridge regression, lasso regression, decision tree regressor and the random forest regressor was used to understand present relationships.

2.2 Research Questions

- To what extent can meteorological and environmental variables explain the variation in forest fire burned areas?
- How does the skewed distribution of the burned area affect the performance of regression model?
- Which regression model provides the most reliable predictions for forest fire damage?
- Do more complex models outperform linear models in predicting forest fire damage?
- Which feature contributes most to predicting the burned area and are these effects strong or limited?

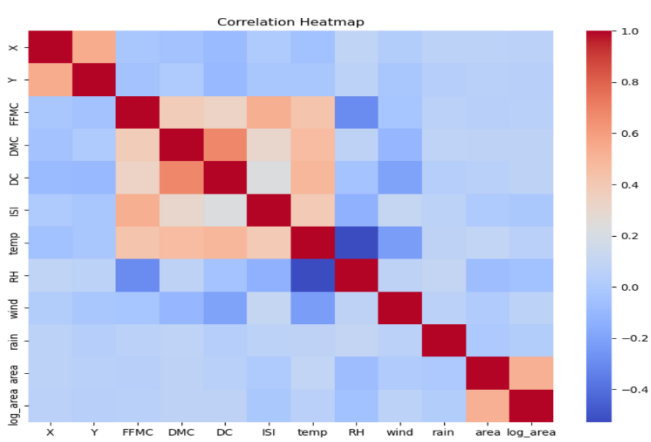
2.3 Exploratory Data Analysis



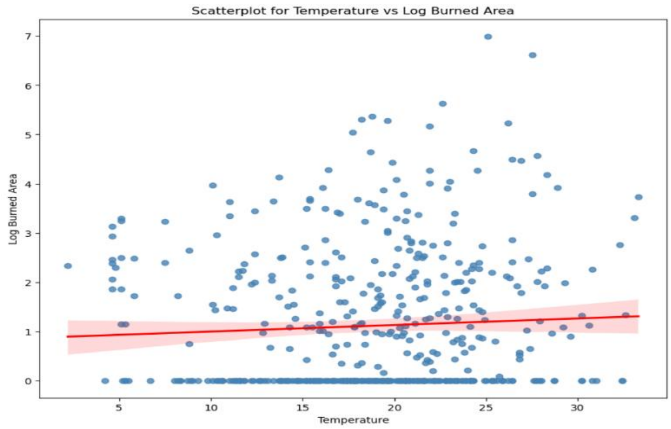
Exploratory data analysis was conducted to understand the distribution of the variables and the relationships between predictors and the target variable. The histograms of the log-transformed burned area show a highly right skewed distribution, with a high concentration of observations close to zero and a long tail of larger values. This indicates that majority of forest fires result in small, burned areas, while a few extreme cases contribute to higher values. The use of the log transformation assisted in reducing skewness; however, the distribution remains slightly uneven.

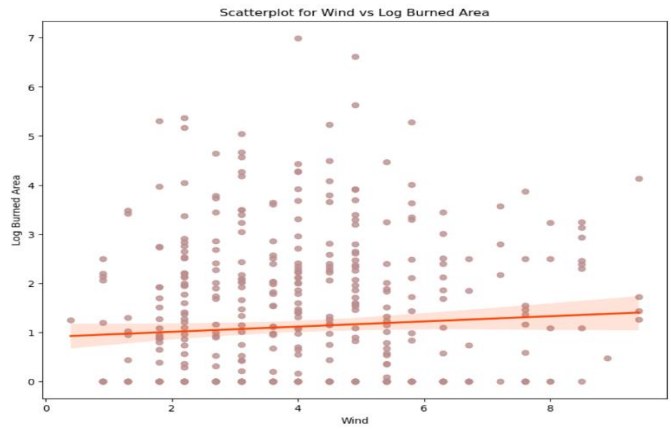
The correlation heatmap outlines generally weak relationships between predictor variables and the burned area. Most correlation coefficients lie close

to zero, indicating that no single variable strongly explains the severity of fire. Moderate correlations are observed between meteorological variables (**temperature, DMC, and DC**), but these do not translate into strong relationships with the target variable.



Scatterplots were generated to evaluate the relationship between temperature and log burned area, as well as the relationship between wind and log burned area. Their relationships show weak positive trends with significant dispersion. The wide spread of points indicates high variability among variables and suggests that these variables alone are not sufficient to accurately predict fire damage.

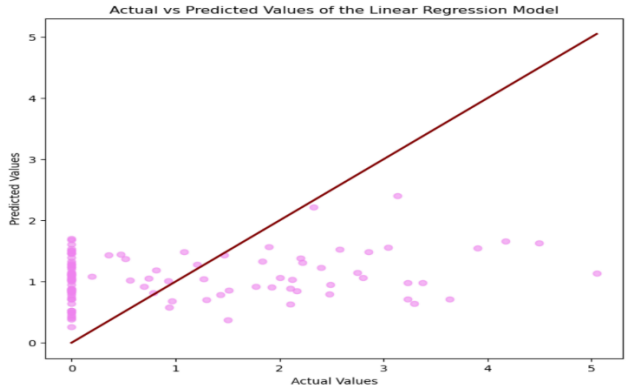




2.4 Regression Models

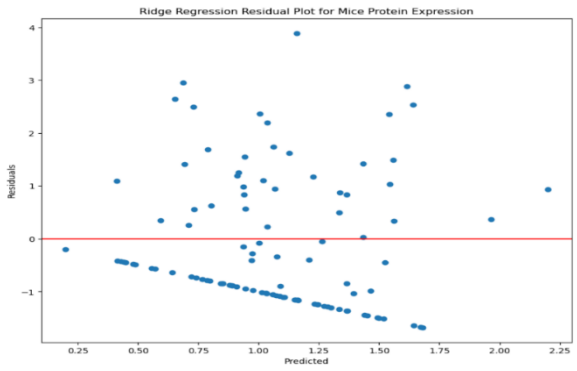
2.4.1 Linear Regression

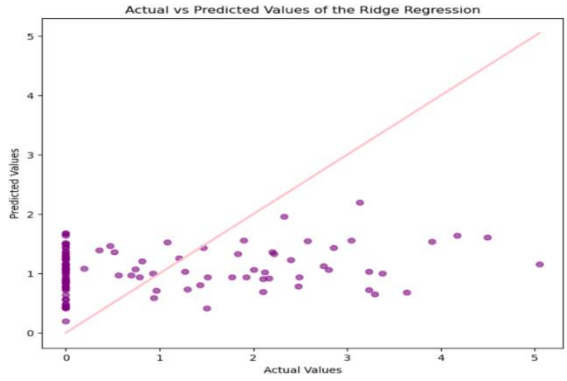
A linear regression model was used as a baseline to model the relationship between the predictor variables and the log transformed burned area. A plot was generated to compare the actual values against those predicted. The plot demonstrated predicted values being concentrated at a narrow range and does not align with the diagonal reference line, indicating that the model struggles to capture variability in data. Furthermore, many observations with actual low values are overestimated while the higher values are underestimated, suggesting that the model is underfitting and unable to represent the complexity of relationships present in the data set.



2.4.2 Ridge Regression

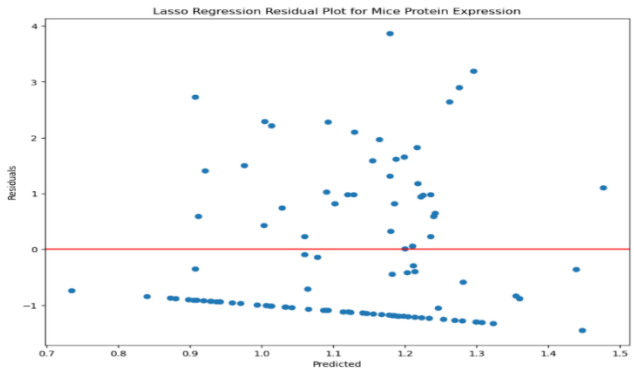
Ridge regression was applied to address the potential existence of multicollinearity among the predictors by applying L2 regularization. An actual vs predicted plot was generated as well and shows that predictions are compressed and fail to capture the spread of data. A residual plot further highlights systematic patterns present rather than random dispersion, highlighting that the model does not fully capture the underlying structure of data. Results generated suggest that ridge regression does not significantly improve predictive performance compared to a baseline model.

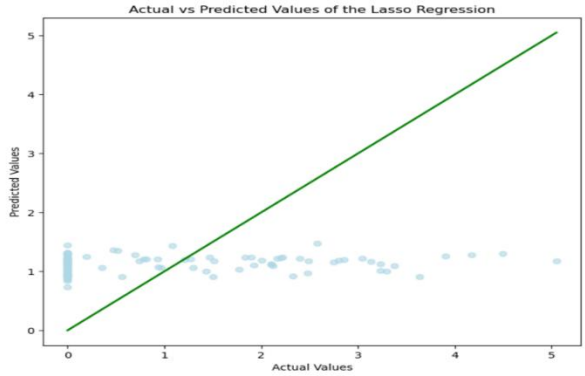




2.4.3 Lasso Regression

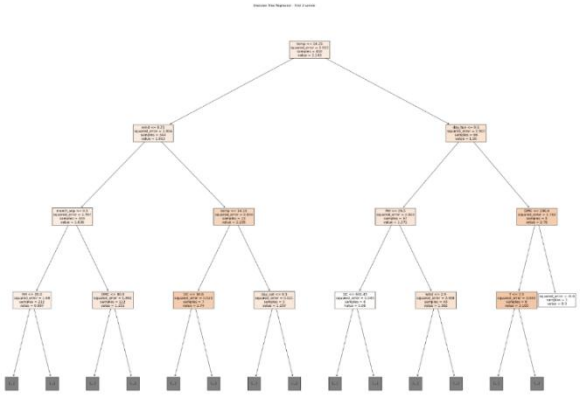
Lasso regression was utilized to perform both regularization and feature selection. The actual vs predicted plot generated highlights that the model outputs highly constrained predictions with most of its values clustered within a narrow range. Such an output suggests that lasso has over penalized the coefficients, effectively simplifying the model too much and causing it to be underfitted. Hence, the model fails to capture meaningful variation in the target variable.

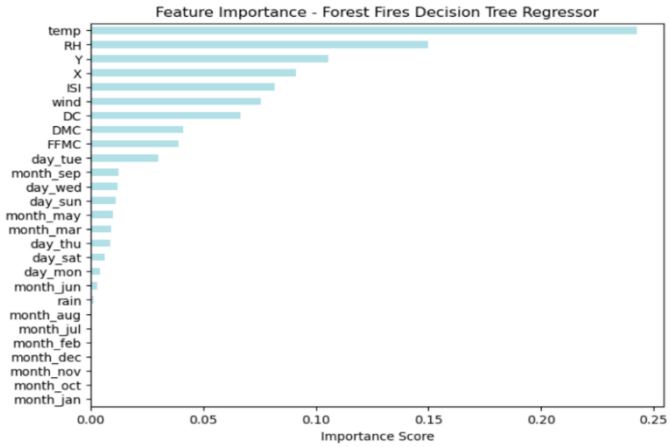




2.4.4 Decision Tree Regressor

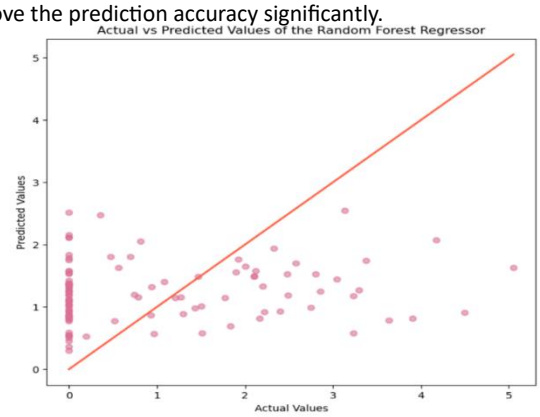
A decision tree regressor was used to capture the possibly existing non-linear relationships in data. An actual vs predicted plot generated for it showed a wide dispersion of predictions, with multiple points deviating significantly from the diagonal reference line. This indicates poor generalization and suggests that the model may be overfitting to the training data while failing to perform well on unseen data. Although the tree structure highlights the splits based on variables (**temperature and relative humidity**), these relationships are not strong enough to generate accurate predictions.

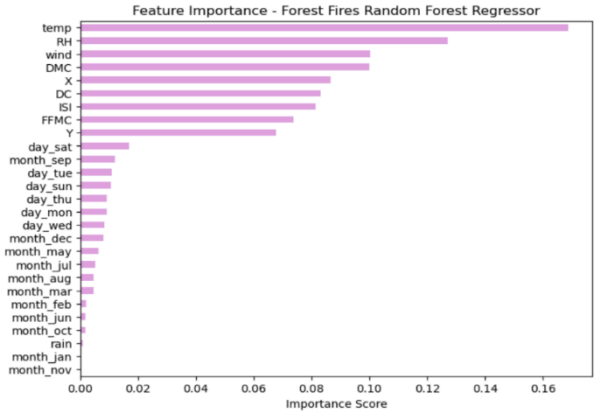




2.4.5 Random Forest Regressor

A random forest regressor was applied to improve predictive performance by combining multiple decision trees. The feature importance plot depicts that temperature and relative humidity are the most influential predictors, followed by wind and drought related indices. However, the actual vs predicted plot generated shows limited alignment with the diagonal reference line indicating that the model struggles to capture the variability of the area burned. Regardless, the regressor is still more robust than a single decision tree but does not improve the prediction accuracy significantly.





2.5 Model Comparison & Cross Validation

	Model	RMSE	MAE	R^2
0	Linear Regression	1.266794	1.074020	0.039381
1	Ridge Regression	1.265197	1.076543	0.041800
2	Lasso Regression	1.281411	1.117285	0.017083
3	Decision Tree	1.952667	1.388800	-1.282422
4	Random Forest	1.336212	1.133315	-0.068785

The performance of all models was evaluated using the RMSE, MSE and R^2 metrics along with cross validation. All models exhibit a very low or negative R^2 values, which indicated that the predictors have a low explanatory power over the burned area. Ridge regression obtained the highest R^2 value (0.0418), even though it remains extremely low. Linear regression performed similarly, while lasso regression output a slightly worse performance due to excessive regularization. The decision tree model
produced a highly negative R^2 value (-1.28), indicating severe overfitting and poor generalization. The random forest regressor showed some improvement however it still resulted in a negative R^2 value. The cross-validation results further confirm these findings; most models result in negative R^2 values which suggest instability and weak predictive capability. Overall, the results indicate that none of the models effectively predicted the area burned by forest fires, likely due to the weak relations among predictors and the target variable, as identified during exploration data analysis.

	Model	CV Mean R^2	CV Std Dev
0	Linear Regression	-0.260869	0.444766
1	Ridge Regression	-0.259327	0.445150
2	Lasso Regression	-0.004421	0.018851
3	Decision Tree	-0.870169	0.274573
4	Random Forest	-0.121250	0.052278

2.6 Conclusion on Regression

The results indicate that all regression models performed poorly in predicting forest fire burned area, with consistently low or negative R^2 values. This aligns with the exploratory data analysis, which showed weak correlations and high variability in the dataset. While linear models performed marginally better than tree-based methods, none were able to capture the underlying patterns effectively. Overall, the findings suggest that the available variables are insufficient to accurately predict fire severity, highlighting the complexity of forest fire behavior.

Task 03 : Classification

3.1 Introduction

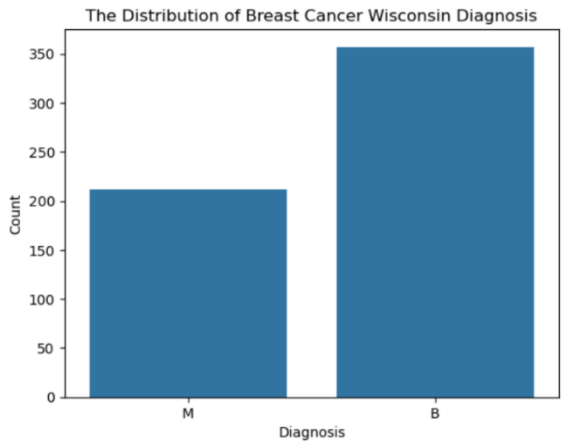
This analysis applies classification techniques to evaluate the Breast Cancer Wisconsin (Diagnostic) dataset obtained from the UCI machine learning repository, to predict whether a tumor is benign or malignant. Exploratory data analysis is conducted to examine the distribution and relationships between the features before model development. Many classification models are further implemented and evaluated to identify the most effective approach for accurate prediction.

3.2 Research Questions

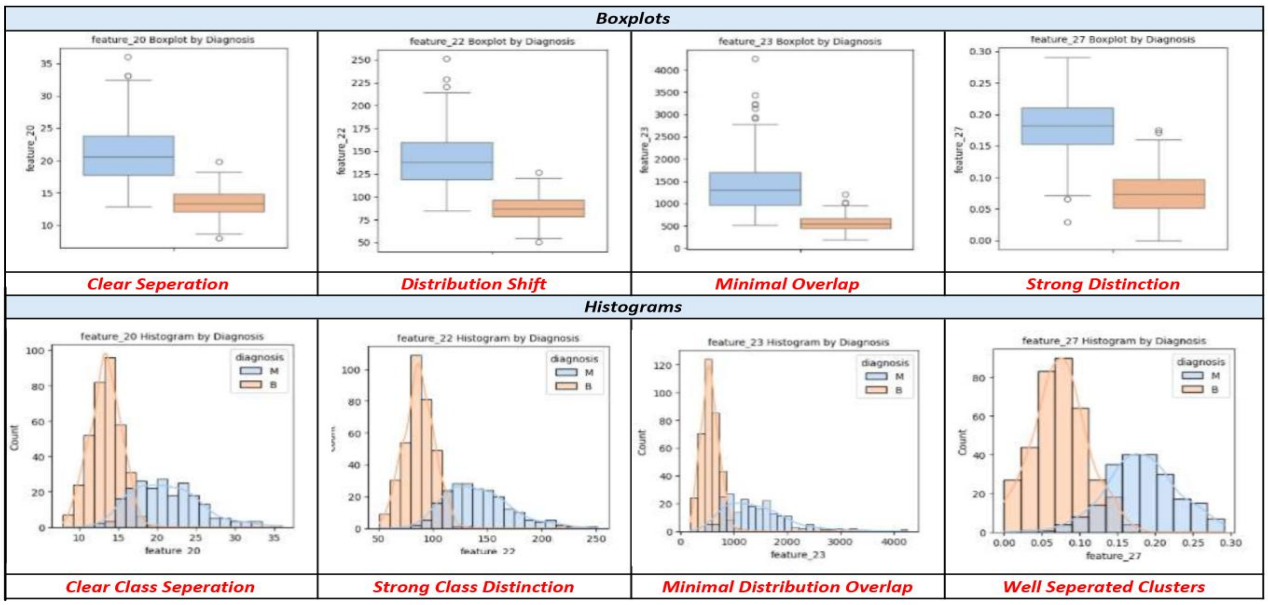
- Which features are most strongly associated with tumor diagnosis?
- Which classification model provides the best predictive performance?
- How accurately can machine learning models classify tumors as benign or malignant using the Breast Cancer Wisconsin dataset?

3.3 Exploratory Data Analysis

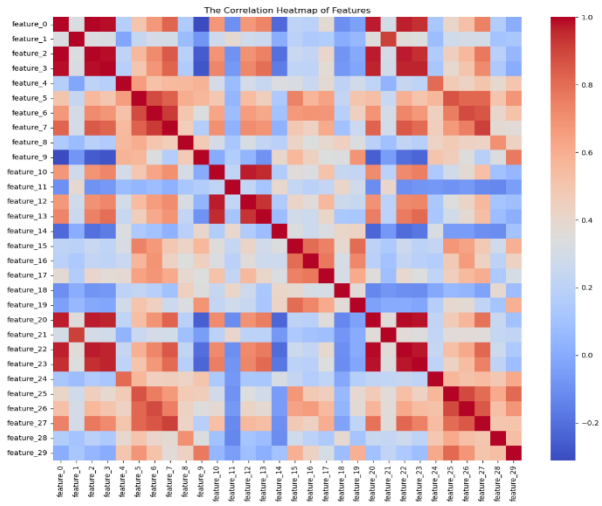
Exploratory Data Analysis was conducted to understand the structure and key characteristics of the dataset. The class distribution plot shows that benign (**B**) cases are more frequent than malignant (**M**) cases, indicating a moderate class imbalance.



Histograms were used to evaluate the distribution of features and reveal any noticeable differences between benign and malignant tumors, suggesting strong discriminatory power. Boxplots further highlight these differences, with malignant cases generally exhibiting higher values and greater variability across several features.



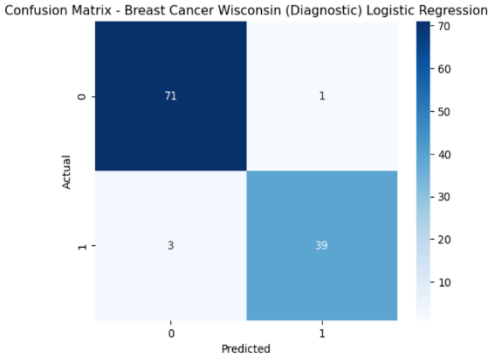
The correlation heatmap indicates strong relationships with multiple features, as well as significant correlations with the diagnosis variable, suggesting that some features are highly informative for classification. In addition, the feature correlation plot identifies the most influential variables associated with the target, providing further insight into feature importance.



3.4 Classification Models

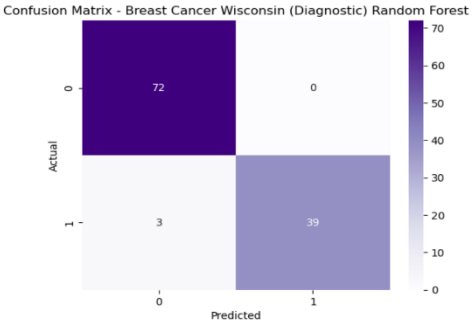
3.4.1 Logistic Regression

Logistic Regression was used as a baseline model due to its simplicity and effectiveness for binary classification problems. The confusion matrix generated classified majority of both benign and malignant cases correctly with a minimal number of misclassifications. There is a low number of false positives and false negatives, indication the model is accurate and reliable in terms of distinguishing between the two classes.



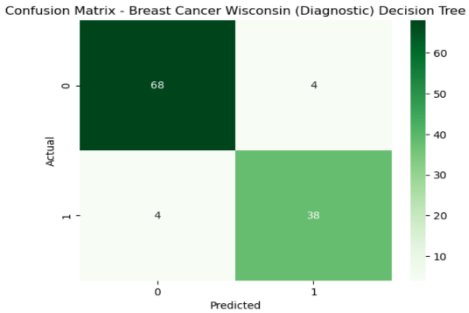
3.4.2 Random Forest Classifier

Random Forest was used as an ensemble method capable of capturing complex, non-linear relationships. The confusion matrix plotted demonstrates an almost perfect classification performance, with almost all observations accurately predicted. There are visible no false positives however, there a very few false negatives, highlighting the model's superior ability to accurately identify the class of the tumor.



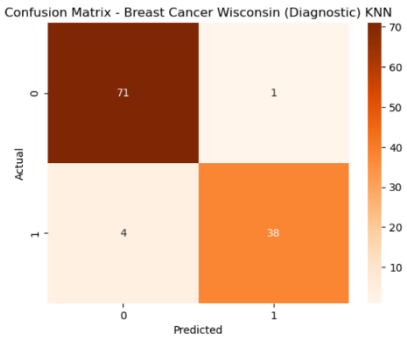
3.4.3 Decision Tree Classifier

Decision Tree was used for its interpretability and ability to model non-linear relationships. The confusion matrix displayed a higher number of misclassifications compared to other models. In specific, there are more false positives and false negatives which suggest the model struggles to generalize well and may be prone to overfitting.



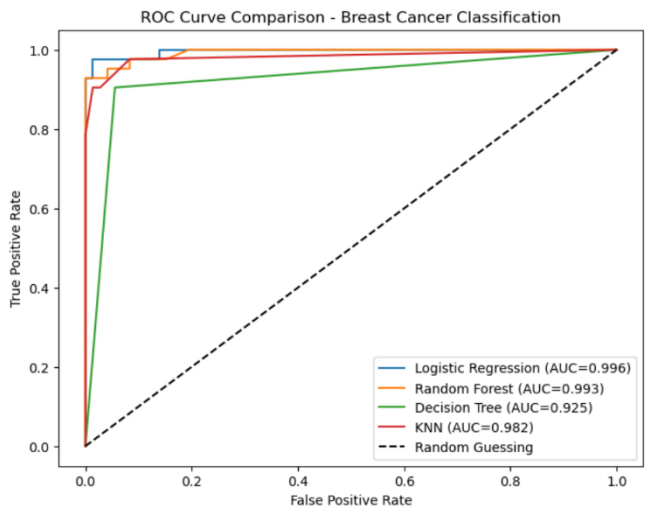
3.4.4 K – Nearest Neighbors (KNN)

KNN was used as a distance-based method that classifies observations based on similarity. The confusion matrix suggests a strong performance with most observations classified correctly. There are slightly more miscalculations compared to logistic regression and the random forest
classifier, specifically in identifying malignant cases, which slightly affects its reliability.



3.5 Model Comparison & Cross-Validation

The ROC curve comparison evaluates the ability of each model to distinguish between benign and malignant tumors. Logistic Regression achieves the highest AUC of **0.996**, indicating excellent classification performance and near-perfect separability between the two classes. Random Forest also performs exceptionally well with an AUC of **0.993**, showing strong predictive capability. KNN demonstrates good performance with an AUC of **0.982**, although slightly lower than the top models, suggesting minor limitations in distinguishing between classes. In contrast, the Decision Tree model has the lowest AUC of **0.925**, indicating comparatively weaker performance.



Random Forest achieved the best performance across most evaluation metrics, including the highest accuracy and precision. This suggests that ensemble methods are highly effective for this dataset, as they combine multiple decision trees to capture complex, non-linear relationships while reducing overfitting. Logistic Regression also performed exceptionally well, indicating that the dataset is largely linearly separable. This implies that the features provide clear boundaries between benign and malignant classes, allowing a simple linear model to perform effectively. The KNN model produced strong results, as it classifies observations based on similarity in feature space. Its good performance suggests that instances of the same class are closely grouped together, although it is slightly less robust compared to ensemble methods. In contrast, the Decision Tree model showed comparatively weaker performance, likely due to overfitting. Decision trees tend to fit the training data too closely, which reduces their ability to generalize to unseen data. The ROC curve comparison further supports these findings, showing that Random Forest and Logistic Regression achieve the highest AUC values, indicating superior classification performance.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.965	0.975	0.929	0.951
Random Forest	0.974	1.000	0.929	0.963
Decision Tree	0.930	0.905	0.905	0.905
KNN	0.956	0.974	0.905	0.938

	Model	AUC
0	Logistic Regression	0.996032
1	Random Forest	0.992890
2	Decision Tree	0.924603
3	KNN	0.982308

Furthermore, cross-validation was performed to assess the stability and generalizability of the models. Logistic Regression achieved the highest mean accuracy with low standard deviation, indicating consistent performance across different data splits. Random Forest also performed well but showed slightly lower stability. Decision Tree exhibited the lowest performance and highest variability, suggesting overfitting.

3.6 Conclusion on Classification

This study shows that machine learning models can effectively classify breast cancer diagnoses with high accuracy. Random Forest achieved the best overall performance due to its ability to capture complex patterns, while Logistic Regression also performed exceptionally well, indicating strong linear separability in the dataset. Evaluation through confusion matrices and ROC curves confirms that both models provide reliable and accurate predictions. In contrast, Decision Tree showed weaker performance due to overfitting, while KNN performed well but slightly below the top models. Overall, the results highlight the importance of model selection and demonstrate the effectiveness of machine learning in supporting accurate medical diagnosis.